



## Problem

Diagnosing WiFi problems in a home can be a pain. It's hard to know exactly where areas of weak connection exist and how to fix them. When a technician attempts to improve a customer's network, they lack a tool that easily diagnoses and solves the issue.

## Implementation

The SmartNetwork Android app uses Iperf3, a cross-compiled C library, for transmitting network data to the Android device. Then, our innovative algorithm uses this information to generate a heatmap displaying areas of strong and weak connectivity. Each heatmap is saved into a database, so it can be loaded at any time using our server with Ruby on Rails.

## Solution



### Speed Test

Discover the exact speed of the WiFi connection at a specific location



### Heatmaps

Generate wifi heatmaps of network connectivity in residences

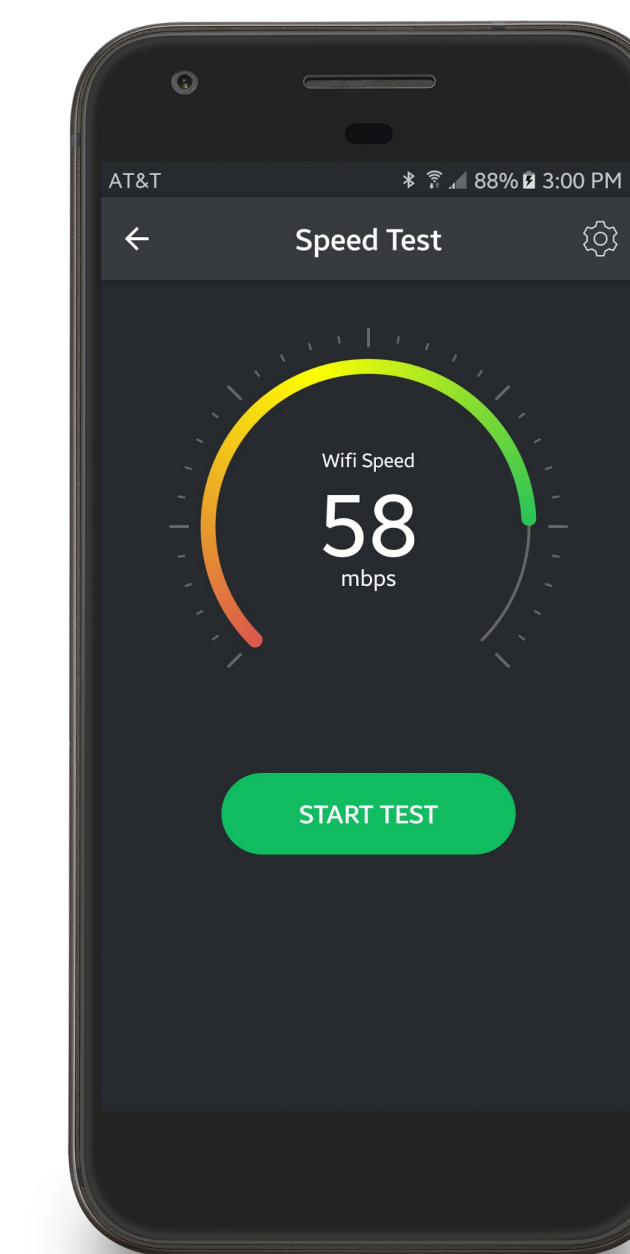


### Persistence

Save and load heatmaps to observe network trends over a period of time



## Design



UC SANTA BARBARA  
engineering

